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Helicopter Landing Zone Criteria

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Purpose

- The purpose of this course is to instruct prospective graduates on required criteria for a Helicopter Landing Zone to be considered usable during Survey and Assault Zone Assessment (SAAZA).



Learning Objectives

- Identify the dimensions of the imaginary surface, to include slope ratio
- Identify the required criteria for an HLZ



References

- UFC 3-260-1 Airfield and Heliport Planning and Design
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures



Outline

- Helicopter Landing Zone (HLZ) Criteria
- HLZ Form 4303
- Review
- Summary



HLZ

- Two common types of HLZ
 - Hasty
 - Located in forward battle areas
 - Requires no construction
 - Pre-planned
 - Construction is limited to only what is required for safety / efficiency in adverse weather conditions
 - Both type HLZs are selected / cleared for minimum landing criteria



HLZ Survey

- Always survey for the maximum size and / or number of Helicopters that the area will support
- All obstacles within the immediate HLZ area are of concern in the HLZ survey
- Give recommended approach / departure axis



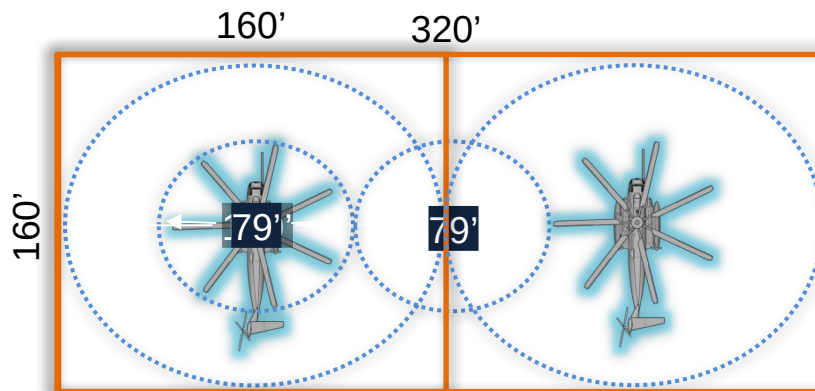
HLZ Survey

- List any quadrants that may restrict approach / departure
- Determine the ground gradients
- Longitudinal gradients are determined from the end of the HLZ that the nose of the aircraft will point. When landing, to the end of the HLZ that the tail will be pointed
- Transverse gradient is from the pilot's right to left



Actual Surfaces

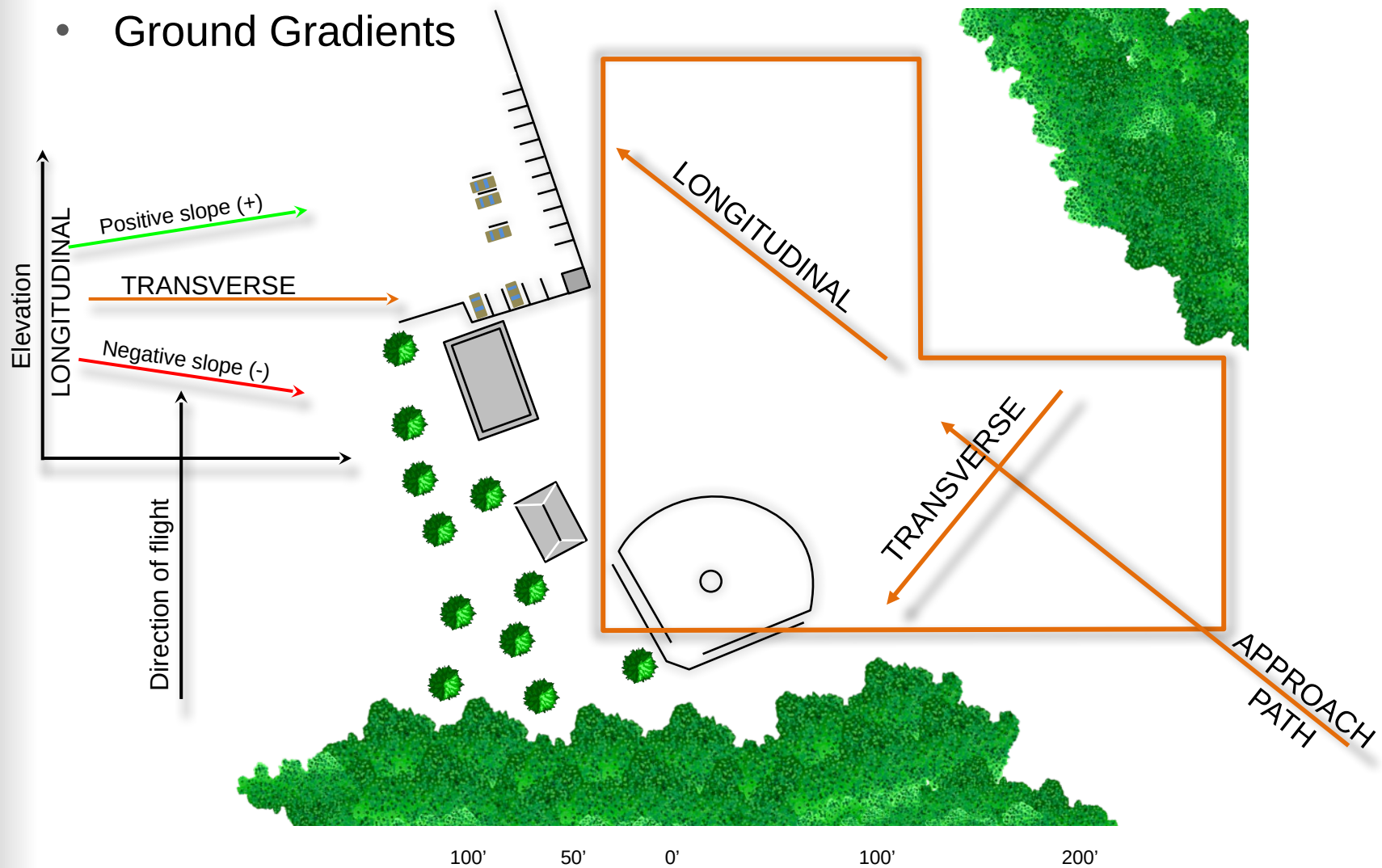
- Criteria is based on number / type
 - Two-rotor blade length spacing is required for blade clearance
 - Single-rotor blade length clearance between each additional helicopter





Actual Surfaces

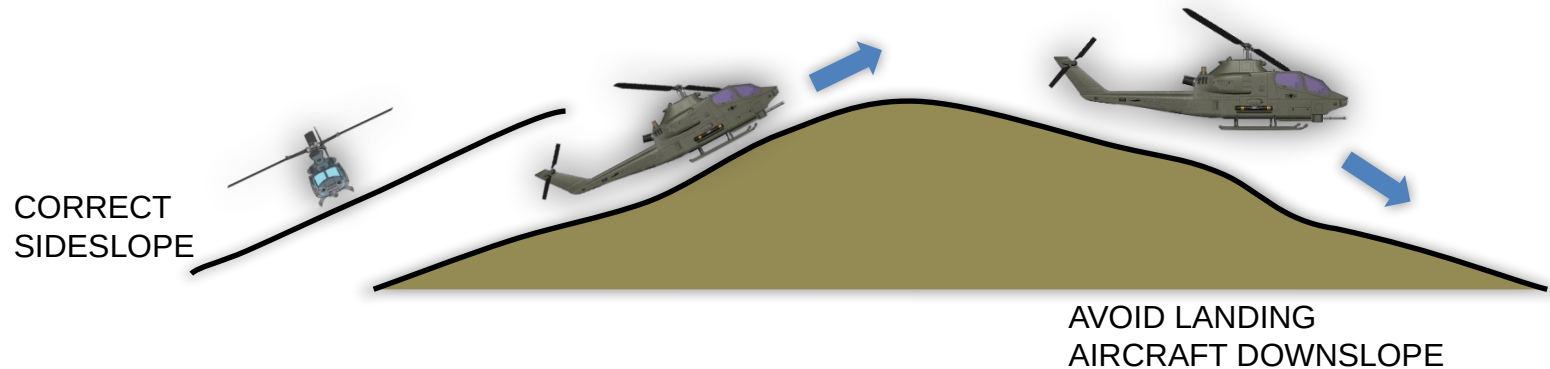
- Ground Gradients





Criteria

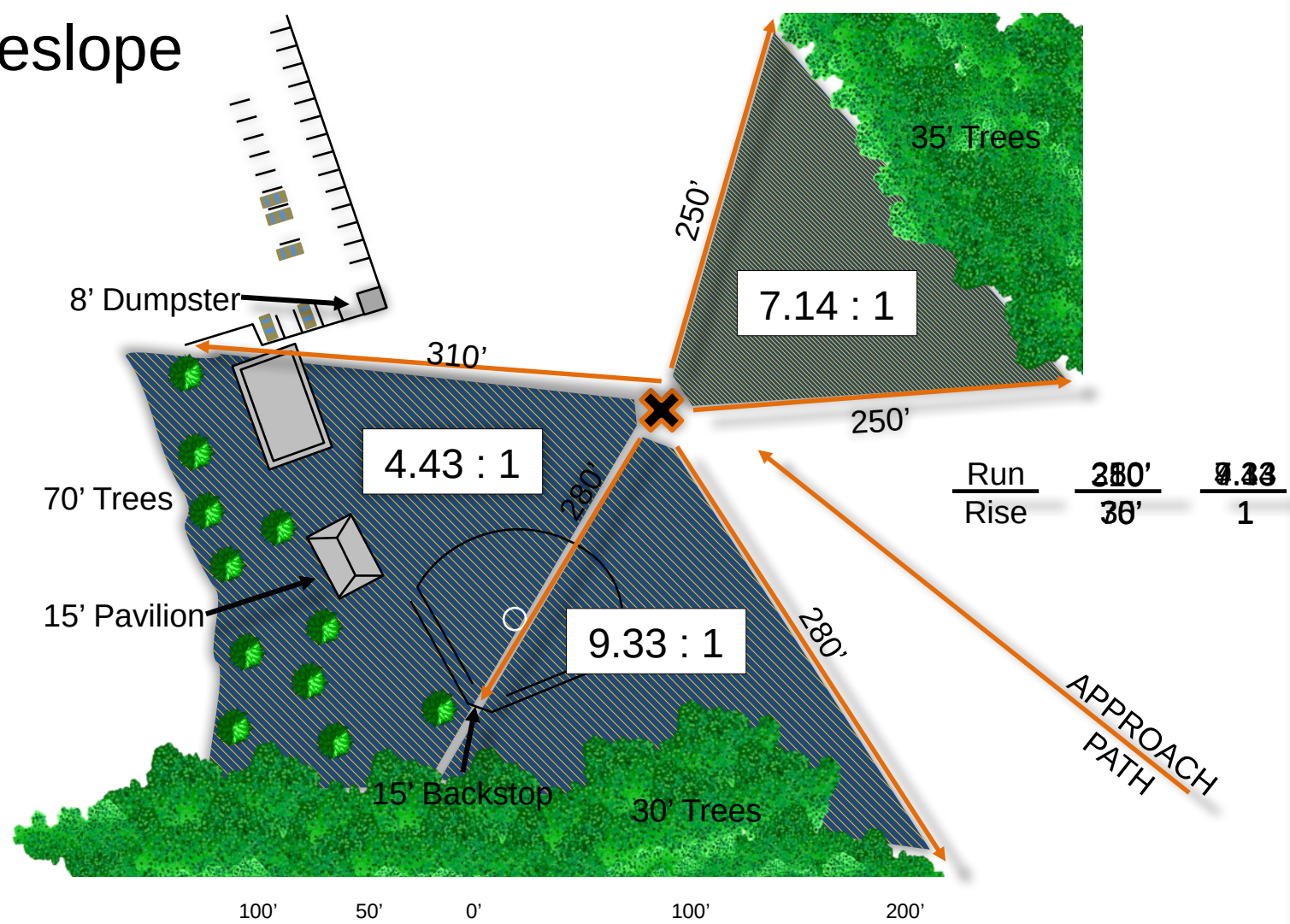
- Avoid slopes exceeding 7°





Imaginary Surfaces

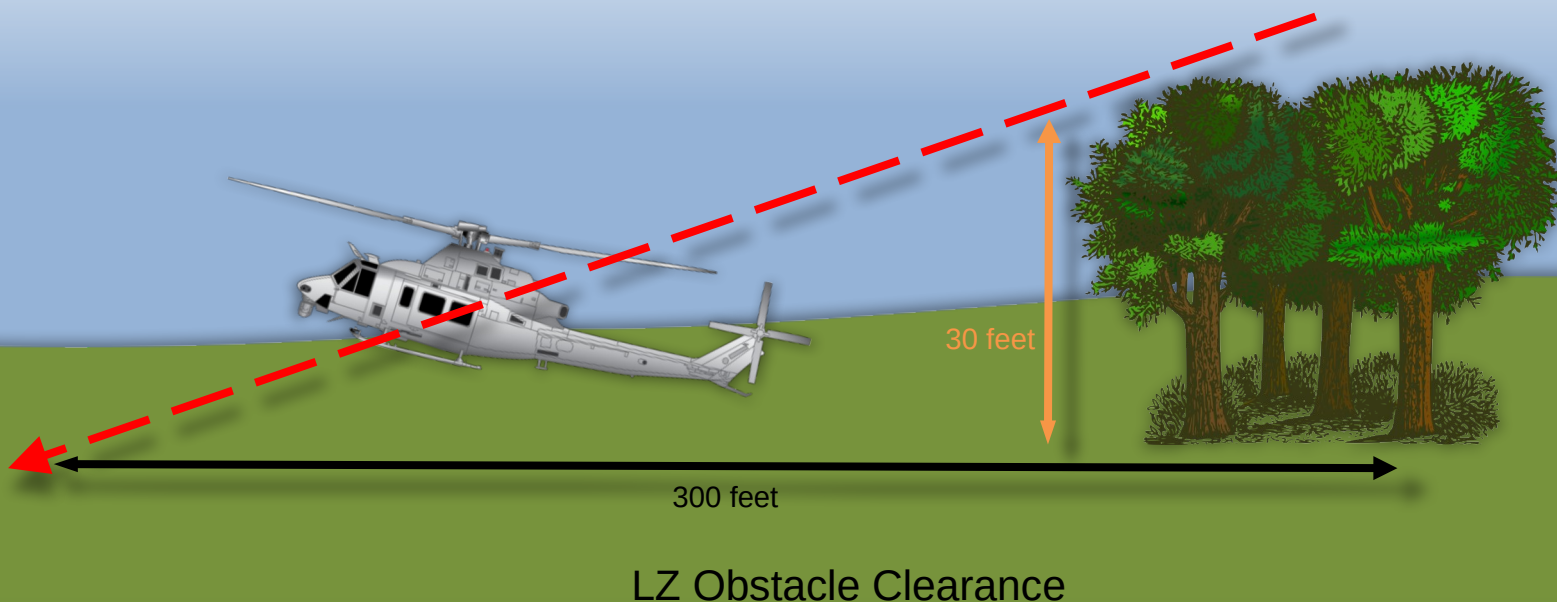
- Glideslope





Imaginary Surfaces

- Landing point should be a distance 10 times as far from highest obstacle 10:1 (5.70)





Outline

- Helicopter Landing Zone (HLZ)
- **HLZ Form 4303**
- Review
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HLZ Form 4303

HELICOPTER LANDING ZONE SURVEY	1A. HLZ NAME	1B. ZAR INDEX NO.	2A. COUNTRY	2B. STATE
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP			
4. SURVEY APPROVAL/DISAPPROVAL DATA				
A. DATE SURVEYED	TYPED NAME AND GRADE OF SURVEYOR		PHONE NUMBER (DSN)	LOCATION
B. DATE REVIEWED	TYPED NAME AND GRADE OF REVIEWER		PHONE NUMBER (DSN)	SIGNATURE
	UNIT AND LOCATION			
C. DATE	TYPED NAME AND GRADE OF APPROVING AUTHORITY		PHONE NUMBER (DSN)	SIGNATURE
APPROVED ☐ DISAPPROVED ☐	UNIT AND LOCATION			
5. COORDINATING ACTIVITIES				
A. HLZ CONTROLLING AGENCY OR UNIT			PHONE NUMBER (DSN)	
B. RANGE CONTROL			PHONE NUMBER (DSN)	
6. HLZ AXIS DATA (APPROACH/DEPARTURE)				
A. MAGNETIC	B. GRID (MGRS)	C. TRUE	D. SOURCE/DATE OF VARIATION DATA	
7. HLZ COORDINATES				
A. SPHEROID/DATUM	B. GPS DERIVED ☐ YES ☐ NO	C. GRID ZONE (52 S PQ)	D. EASTING	E. NORTHING
F. HLZ CENTER - POINT	MGRS COORDINATES	WGS84 LATITUDE (D-M-M-M)		WGS84 LONGITUDE (D-M-M-M)
8. HLZ SURFACE DATA				
A. LENGTH (FEET)	B. WIDTH (FEET)	C. ELEVATION	D. QTY/TYPER(2H-53: 2H-50)	
E. QUADRANT (OBSTRUCTED/UNOBSTRUCTED)		F. SLOPE		
9. REMARKS				

HLZ NAME	
10. HLZ DIA/GRAM	
11. PHOTOGRAPHY AVAILABLE	
☐ YES ☐ NO	
LOW LEVEL ROUTES	
☐ NONE AVAILABLE	
☐ ROUTE NAME/DESIGNATOR	



HLZ Form 4303

- Blocks 1-3 Header

HELICOPTER LANDING ZONE SURVEY	1A. HLZ NAME	1B. ZAR INDEX NO.	2A. COUNTRY	2B. STATE
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP			



HLZ Form 4303

- Block 4 Routing

4. SURVEY APPROVAL/DISAPPROVAL DATA			
A. DATE SURVEYED	TYPED NAME AND GRADE OF SURVEYOR	PHONE NUMBER (DSN)	LOCATION
B. DATE REVIEWED	TYPED NAME AND GRADE OF REVIEWER	PHONE NUMBER (DSN)	SIGNATURE
	UNIT AND LOCATION		
C. DATE	TYPED NAME AND GRADE OF APPROVING AUTHORITY	PHONE NUMBER (DSN)	SIGNATURE
APPROVED ☐ DISAPPROVED ☐	UNIT AND LOCATION		



HLZ Form 4303

- Block 5 Coordinating Activities

5. COORDINATING ACTIVITIES	
A. HLZ CONTROLLING AGENCY OR UNIT	PHONE NUMBER (DSN)
B. RANGE CONTROL	PHONE NUMBER (DSN)



HLZ Form 4303

- Block 6 Approach and Departure

6. HLZ AXIS DATA (APPROACH/DEPARTURE)			
A. MAGNETIC	B. GRID (<i>MGRS</i>)	C. TRUE	D. SOURCE/DATE OF VARIATION DATA



HLZ Form 4303

- Block 7 Coordinates

7. HLZ COORDINATES				
A. SPHEROID/DATUM	B. GPS DERIVED ☐ YES ☐ NO	C. GRID ZONE (52 S PQ)	D. EASTING	E. NORTHING
F. HLZ CENTER - POINT	MGRS COORDINATES	WGS84 LATITUDE (D-M.MM)	WGS84 LONGITUDE (D-M.MM)	



HLZ Form 4303

- Block 8 HLZ Surface Data

8. HLZ SURFACE DATA			
A. LENGTH (FEET)	B. WIDTH (FEET)	C. ELEVATION	D. QTY/TYPE (2/H-53; 2/H-60)
E. QUADRANT (OBSTRUCTED/UNOBSTRUCTED)		F. SLOPE	



HLZ Form 4303

- Block 9 Remarks

9.	REMARKS
	SURFACE: Example: Grass/BROWNOUT-WHITEOUT: No Factor or CAUTION: BROWNOUT-WHITEOUT CONDITIONS MAY EXIST! Additional HLZ recommended Approach/Departure Headings: APP xxx.x °/ DEP xxx.x ° - Coordinates and elevation data derived from GPS, maintained horizontal margin of error +/- "xx" meters, (use least accurate when collecting data) Vertical margin of error 1.5 X horizontal error. MAGVAR = "x.x"°E/W/changing "x.x"°E/W per year/derived from "WMM 20xx" as of date surveyed. - Administration/Coordination/Planning/Scheduling Instructions: - Surveyor Observed Obstacles/Hazards on and within 1,000 meters of the HLZ Boundary (All headings magnetic and distances from HLZ Center point unless otherwise specified). User is responsible for confirming ALL obstacle/hazard locations prior to conducting HLZ operations: <u>Obstacles/Hazards: INSIDE HLZ:</u> a. e.g., 25' Windsock 15' @ 225° from HLZ Center point. . <u>Obstacles/Hazards: OUTSIDE HLZ:</u> a. e.g., 40' Trees border East, South and West borders of HLZ.



HLZ Form 4303

- Block 9 (continued)

Quadrant Controlling Obstacle: GSR 6:1 or less = Obstructed/GSR 7:1 or more Unobstructed.
(Headings 90° or Cardinal/Obstructed or Unobstructed/Obstacle Height/Distance from HLZ Cntr/GSR)
xxx°-xxx° or Cardinal/xxx' Obstacle Name/xxxx' from HLZ Center/ GSR xx:1
xxx°-xxx° or Cardinal/xxx' Obstacle Name/xxxx' from HLZ Center/ GSR xx:1
xxx°-xxx° or Cardinal/xxx' Obstacle Name/xxxx' from HLZ Center/ GSR xx:1
xxx°-xxx° or Cardinal/xxx' Obstacle Name/xxxx' from HLZ Center/ GSR xx:1

4. Survey or Observed Terrain Obstacles/Hazards:

a. e.g., hilltop est. 2 NM north, elev. unknown.

5. Safety of Flight Review:

a. SOF Review Reported Obstacles/Hazards (including terrain) within 5 NM Radius of HLZ Center:

i. North: e.g., multiple towers, buildings, high terrain and smokestacks.

ii. South: e.g., rapidly rising terrain @ 0.2 NM,

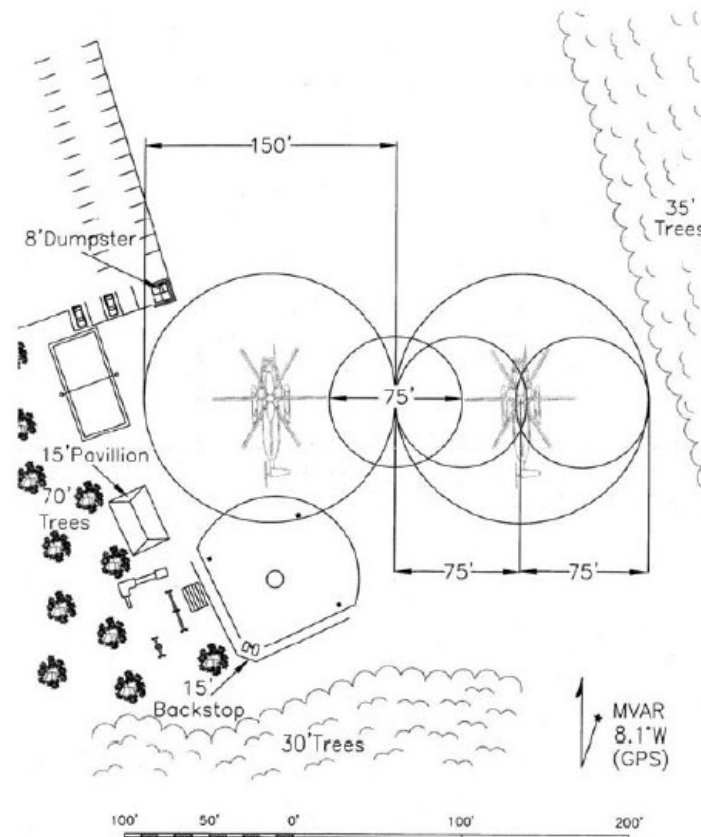
b. Additional Information and Airspace:

i. xxx are located within a 10 NM Radius of the HLZ Center point.

6. Notes:



- Block 10 HLZ Diagram





Outline

- Helicopter Landing Zone (HLZ)
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- **Review**
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Review

- What is the minimum glideslope for all helicopters?
 - 10:1
- What is the maximum safe landing slope for a helicopter?
 - 7 degrees
- What is the name of a HLZ survey form?
 - 4303
- What are the two common types of HLZ surveys?
 - Hasty
 - Pre-planned



Outline

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Summary

- Helicopter Landing Zone (HLZ)
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Questions



Student Survey



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